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**ATOMIC INSIGHTS ON INTERACTION MECHANISM OF DISLOCATION WITH VOID/IMPURITY/  
PRECIPITATES IN BCC IRON**

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**ABSTRACT**

*Ferritic steels are important cladding and structural material for current and advanced reactors containing Fe as a primary compositional matrix element. Three proposed models containing voids, impurity P, and nano oxides embedded in Fe were simulated using molecular dynamics code LAMMPS.*

*The interaction mechanism of dislocation for each model was observed and compared by analyzing its strengthening effect through the stress-strain curve. In considering the interaction of dislocation with voids, it was observed that for all void sizes, dislocation bypass voids following orowan unpinning mechanism without any loop irrespective of any radii, while for P clusters, dislocation is trapped at the interface and leaves it after making a screw dipole shape following orowan mechanism. Moreover, the oxide dislocation interaction in oxide dispersed in iron follow the orowan unpinning mechanism accompanied by loops around oxide. The values of critical unpinning stress for all models were analyzed and compared. Atomic insights on the interaction mechanism indicate that the interaction mechanism is essentially the same for void and a P cluster in Fe i.e. orowan unpinning mechanism. The presence of oxide in Fe matrix during its interaction follows the orowan unpinning mechanism along with a dislocation loop around the oxide and its area is proportional to the size of the oxide. The critical unpinning stress was compared and analyzed for each model.*

*This work will provide a unique and valuable insight into the strengthening mechanism due to dispersed oxides by providing new parameters for the multi-scale simulation. Our results demonstrate that oxide impedes dislocations and offers hindrance for edge dislocations to saturate further. Furthermore, our simulation results provide atomic insight into the onset of plasticity in iron alloys.*

**Keywords:** Molecular dynamics; Ferritic steels, edge dislocation; orowan loop; oxide dispersed strengthened steel.

**1. INTRODUCTION**

The structural components in radiation environments, especially exposed to fast neutron, create radiation damage effects mostly in the form of vacancies-interstitials pairs which further extend into voids and other radiation debris [1]. The existence of different types of defects together with alloying elements, impurities/voids/precipitates affect the movement of the edge dislocation in iron. In irradiated environments, a combination of multiple voids together with dislocations exists and affects the material behavior [5]. The defects impede the dislocation motion, thereby results in hardening and loss of ductility. The strength of structural materials attributed to the presence of voids and precipitates that can constrain the mobility of the dislocations [2].

The study of defects and their interactions is important for designing structural materials for reactors. The void-defects interaction in Fe and their study is very important because their extension can result in radiation-induced hardening and related issues.

The interaction mechanism and dependence of phosphorus concentration on the strengthening effect are studied in the second model. It is pertinent to mention here that P is an impurity that is considered as an essential constituent in steel that can effectively improve the strength of steel [3–5]. In addition to enhancing strength, phosphorus also has a vital role in improving wear resistance and corrosion resistance. The Fe-P binary alloys containing an impurity concentration of more than 0.05% can enhance GB strengths [6]. Another research reported that concentration in the range of 0.3-0.7% in iron under deformation could predominantly increase mechanical strength [7]. Additionally, P is considered an important solid strengthener of steels by trapping defects at the GBs [8]. The addition of only up to 0.2 at.% increases both the yield and tensile strength of steel along with improved